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Sebastian Erfort,[1,a)] Martin Tschöpe,[1] Guntram Rauhut,[1,b)] Xiaoqing Zeng,[2] and David P. Tew[3,c)]

AFFILIATIONS

[1] Institute for Theoretical Chemistry, University of Stuttgart, Pfaffenwaldring 55, 70569 Stuttgart, Germany
[2] Department of Chemistry, Fudan University, Shanghai 200433, China
[3] Max Planck Institute for Solid State Research, Heisenbergstr. 1, 70569 Stuttgart, Germany

[a)] erfort@theochem.uni-stuttgart.de
[b)] **Author to whom correspondence should be addressed:** rauhut@theochem.uni-stuttgart.de
[c)] d.tew@fkf.mpg.de

ABSTRACT

The rovibrational spectra of metaphosphorous acid, HOPO, and its deuterated isotopologue have been studied by vibrational configuration interaction calculations, relying on the internal coordinate path Hamiltonian and the Watson Hamiltonian. Tunneling effects for the overtones of the torsional mode, which gives rise to the *cis–trans* isomerization, and its rovibrational transitions have been investigated in detail. Due to strong matrix effects, comparison with experimental data is hindered, and thus, the calculations provide accurate estimates for the fundamental modes of these species.

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I. INTRODUCTION

Nitrous acid, HONO, has been the subject of very many experimental and theoretical studies, and there is an onerous number of articles concerning this molecule, discussing its *cis–trans* isomerization, the 1,3-hydrogen shift, the (un)importance of tunneling, its vibrational spectrum, and many other aspects. For example, Richter *et al.*[1,2] and Luckhaus[3,4] independently performed full-dimensional quantum chemical calculations in order to understand the dynamics of this molecule and its implications for chemical reactions in the atmosphere, while Guilmot and co-workers[5,6] recorded the high-resolution FTIR spectra of both the *cis* and *trans* conformers. Conversely, the closely related metaphosphorous acid, HOPO, has received very little attention.[7–9] Gas phase and matrix isolation experiments have been performed to identify the fundamental modes of *cis*-HOPO, accompanied by density functional calculations using the harmonic approximation.[9] Importantly, Davies and co-workers clearly assigned two fundamental modes in the gas phase to *cis*-HOPO. Although quantum chemical calculations predict that both the *cis* and *trans* conformers are minima on the Born–Oppenheimer potential surface, the higher energy *trans* conformer has only very recently been identified experimentally. Our matrix isolation experiments in solid N_2 proved for the first time the existence of *trans*-HOPO and indicated large matrix effects with respect to the fundamental modes of both conformers.[10] Due to the low-lying transition state of the *cis*–trans isomerization, tunneling effects can, in principle, affect the vibrational dynamics, making accurate calculation of the spectrum a non-trivial task. According to explicitly correlated coupled-cluster calculations, CCSD(T)-F12a/cc-pVTZ-F12, the minimum of the *trans* conformer is 918 cm^{-1} above the *cis* conformer and the torsional barrier is 3241 cm^{-1}. It is therefore probable that low-lying vibrational states of the *trans* conformer interact with high-lying states of the *cis* conformer in a region of high state density. This will affect states associated with the torsional out-of-plane (OOP) mode and those with non-negligible contributions from this mode, as has been noticed by Richter *et al.*[1,2] for HONO.

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In order to study the vibrational spectra of both conformers, two independent types of vibrational configuration interaction calculations (VCI) have been performed. The first one is based on the internal coordinate path Hamiltonian (ICPH) and treats both minima in the same calculation using the minimum energy path for the *cis*–*trans* isomerization. This approach directly accounts for tunneling effects and is well suited for describing this system. The second approach is based on separate calculations for the *cis* and *trans* minima using normal coordinates of each conformer and the Watson Hamiltonian (WH) and thus cannot account for tunneling effects. However, at least for the low-lying vibrational states of the *cis* conformer, which cannot be affected by tunneling, both methods are expected to provide the same state energies at convergence with respect to the VCI parameters and representation of the potential. In this study, we have also investigated the out-of-plane mode in detail by calculating its rovibrational spectrum up to high J-values and by the calculations of the vibrational overtones of this torsional mode. Note that O'Sullivan *et al.* measured rovibrational bands for the ν_2 mode at about 1258 cm^{-1}, which they assigned to J-values of up to 30.[9]

II. THEORY

A. The internal coordinate VSCF/VCI approach

The internal coordinate path Hamiltonian[11,12] treats both the *cis* and *trans* minima in the double well torsional potential on an equal footing. Following the prescription of Hougen, Bunker, and Johns,[13,14] the full 360° torsional motion is treated through a special path in configuration space, and the coordinates for the remaining modes are normal coordinates orthogonal to the path, where these coordinates vary with torsional angle τ. In this work, the torsion coordinate is the minimum energy path $\mathbf{a}(\tau)$ obtained by relaxing other degrees of freedom for fixed angle τ. As pointed out in Ref. 15, the correct choice of axis frame for the periodic path is the inertial frame, in which the moment of inertia at each point along the path is diagonal. In this work, the normal coordinates for the non-torsional modes are defined adiabatically and, at each τ, diagonalize the projected mass-weighted Hessian matrix. Modes ν_3 and ν_4 undergo an avoided crossing in the region of the *trans* minimum where the adiabatic coordinates mix and exchange character. To remove the artificially strong coupling with the torsional coordinate, diabatic coordinates were constructed for these modes by interpolating through the crossing region.

The vibrational self-consistent field (VSCF) wavefunction is an optimized product of modal functions, where a basis of trigonometric functions is used for the torsion and a basis of harmonic oscillator functions for the other modes. The configurations used in the VCI calculation are product wavefunctions formed from the VSCF modal functions resulting from optimizing the VSCF zero point energy. In the DYNAMOL program, the Hamiltonian matrix elements are approximated using a many-body expansion in the normal coordinates, where the torsion is included in every term, and the terms are evaluated by quadrature. For further details of the VSCF and VCI implementations, we refer readers to Ref. 16, and for details of the internal coordinate path Hamiltonian, the Jacobian, and how VCI matrix elements are evaluated using this Hamiltonian, we refer the readers to Ref. 15.

B. The normal coordinate VSCF/VCI approach

Standard VSCF/VCI theory uses normal coordinates of a single reference geometry, typically a stationary point, the Watson Hamiltonian, and has been described extensively in the literature.[17–19] In our calculations, we used the equilibrium structures of the *cis* and *trans* conformers. The PES, which is represented through an n-mode expansion in terms of normal coordinates, is restricted to a single minimum rather than covering both. As the isomerization barrier is not very high in energy, this approach must inevitably lead to problems in the high energy regions, which due to fitting of the grid points to analytical functions (polynomials or B-splines) rely on extrapolation. Usually, this problem has little impact on low-lying fundamental transitions or modes orthogonal to the barrier but may result in instabilities arising from ill-defined correlation functions. In the underlying VSCF calculations, we determine state-specific modals obtained from a linear combination of distributed Gaussian functions, which will be used in a configuration-selective VCI approach based on the Watson Hamiltonian, as described in detail elsewhere.[20]

III. COMPUTATIONAL DETAILS

A. Electronic structure calculations

Electronic structure calculations have been performed at the level of explicitly correlated coupled-cluster theory including single and double excitations and a perturbative treatment of triple excitations, i.e., CCSD(T)-F12a, in combination with an associated basis set of triple-ζ quality, cc-pVTZ-F12. The first step in the explicitly correlated coupled-cluster calculations, i.e., the MP2-F12 calculations, was performed using the MP2-F12/**3**C(FIX) approach. The aug-cc-pVTZ/MP2FIT basis set of Weigend *et al.*[21] was used as auxiliary basis. For evaluating the Fock matrix, the cc-pVTZ/JKFIT basis set[21] was used. Unless otherwise noted, the same basis sets were also used for the resolution of identity (RI). The complementary auxiliary basis set (CABS) approach was employed, i.e., the union of the AO and RI basis sets was used to approximate the resolution of the identity. The Hartree–Fock energies were corrected by addition of the CABS singles correction. Multireference effects were found to be very small in all important regions of the PES.

The two minima and the connecting transition state (showing an imaginary frequency of i548 cm^{-1}) were optimized at this level of theory. Geometrical parameters are provided in the supplementary material. Similar to harmonic frequencies, normal coordinates and all single-point calculations representing the potential have been computed at the same level of theory. All calculations have been performed with the MOLPRO package of *ab initio* programs,[22] but very tight thresholds beyond the default settings have been used in order to reduce numerical noise as much as possible.

B. Potential

For the internal coordinate path VCI calculations using DYNAMOL, a full six-dimensional analytic potential function was fit to CCSD(T)-F12a energies. The data used for the fit were energies and geometries from random displacements in internal coordinates from points along the minimum energy path, resulting in 4707 data

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points in the energy range of 0–20 000 cm^{-1} relative to the *cis* global minimum. The analytic function used to fit the data was

$$V(\mathbf{r}) = V_0(\mathbf{r}) + V_c(\mathbf{r}), \tag{1}$$

$$V_0(\mathbf{r}) = \sum_i D_i^2\left(1 - e^{-\alpha_i(r_i - e_i)}\right)^2, \tag{2}$$

$$V_c(\mathbf{r}) = \sum_i \sum_{\mu_i} d_i^\mu \left(e^{-\beta_i^\mu (r_i - c_i^\mu)^2} - s_i^\mu\right) + \sum_{i>j} \sum_{\nu_{ij}} d_{ij}^\nu \left(e^{-\beta_i^\nu (r_i - c_i^\nu)^2 - \beta_j^\nu (r_j - c_j^\nu)^2} - s_{ij}^\nu\right) + \cdots, \tag{3}$$

where r_i are atom–atom separations and up to four-body terms are included. The Morse parameters of the zeroth-order surface V_0 are first optimized to loosely fit the data with the primary role of ensuring that the potential behaves properly in the asymptotic regions of dissociation and atom–atom coalescence. For the correction surface V_c, the centers c of the multivariate Gaussian functions lie on grids with spacings between 0.1 and 1.0 a_0 and the exponents β were selected by setting the standard deviation to twice the spacing between neighboring Gaussians. The shift parameters s are defined by standardization of the weighted least squares problem. The linear parameters d are determined by a method[16] similar to LASSO regression,[23] where the R^2 quality of the fit is optimized, subject to a constraint on the L1 norm of the vector of coefficients d. In this method, only the most significant basis functions have non-zero coefficients, which produces a very compact analytic function and minimizes the risk of overfitting. A total of 19 033 Gaussian functions were provided to the fitting procedure, which resulted in an accurate analytic function with only 600 functions. A Fortran subroutine for the analytic potential is provided in the supplementary material.

Within the single minimum normal coordinate calculations, an n-mode expansion[24] of the potential in terms of normal coordinates including up to 4-mode coupling terms has been used throughout. Tests with 5-mode coupling terms showed that these do not provide any sizable corrections to the final results. The generation of the potential has been performed in a fully automated manner, exploiting the symmetry within the individual subsurfaces and the electronic structure calculations. For details, see Refs. 25 and 26. An analytical representation of the PES has been obtained from this grid-based one by using an efficient Kronecker product fitting to polynomials up to eighth order along each coordinate.[27] For the calculation of overtones, a representation of the PES in terms of B-splines has been used in order to describe the outer regions of the potential more reliably.

C. Vibrational structure

In the WH-VCI calculations using MOLPRO, the vibrational wavefunction for each state was first optimized at the VSCF level using a basis of 20 distributed Gaussians and neglecting the vibrational angular momentum (VAM) terms[28] in the Watson Hamiltonian.[29] The subsequent WH-VCI calculations used these state-specific modals with a correlation space including up to full six-tuple excitations, a maximal excitation per mode up to the eighth root, and a maximum sum of quantum numbers of 20, which results in 65 536 configurations for A′ states and 56 460 configurations for A″ states. For the calculation of overtones, the latter two parameters were altered to the ninth root and a sum of 30 in combination with modals obtained from 24 basis functions. All WH-VCI calculations included the VAM terms, but with a constant μ-tensor,[28] and used a configuration-selective algorithm[20] that fully exploits symmetry. Eigenvalues were determined with our residuum based eigenvalue solver (RACE).[30]

In the ICPH-VCI calculations using DYNAMOL, the vibrational wavefunction for the zero point was first optimized at the VSCF level using a basis of 12 harmonic oscillator functions for each vibrational mode, with frequencies characteristic of the *cis* minimum, and 45 trigonometric functions for the torsion. The ICPH-VCI calculations were performed using a maximal excitation per vibrational and torsional mode up to the ninth and fifteenth roots, respectively. No restriction is made on the VCI excitation level, but the maximal sum of quanta was restricted to 18 or 14 for excitations that do or do not include the torsional mode, respectively, which was sufficient to converge the states of interest to better than 1 wavenumber. Zero point energies and fundamental and overtone transitions for both the *cis* and *trans* rotamers were obtained from the eigenstate stack resulting from a single ICPH-VCI calculation. The full internal coordinate path Hamiltonian with all VAM terms was used in all ICPH-VSCF and ICPH-VCI calculations, approximated with a many-body expansion including up to fifth-body coupling.

D. Rovibrational structure

The option to calculate rovibrational spectra is a newly developed feature within the vibrational code of the MOLPRO program package.[22] All quantities needed such as *ab initio* potential energy and dipole surfaces, rovibrational energies, nuclear spin statistical weights, and transition moments are determined within the program in an automated fashion to keep the user-input to a bare minimum. A detailed description of the different quantities needed for the spectra and the way they are calculated is given elsewhere,[31,32] and only a short summary is given here.

The rovibrational infrared intensities are computed as[33]

$$I_{\mathrm{if}} = \frac{2\pi^2}{3} \frac{N_{\mathrm{A}}}{\epsilon_0 h^2 c^2} g_{\mathrm{ns}} \frac{e^{-E_{\mathrm{i}}/kT}\left(1 - e^{-\nu_{\mathrm{if}}/kT}\right)}{Q(T)} \nu_{if} S_{\mathrm{if}}, \tag{4}$$

with natural constants N_{A}, ϵ_0, h, and c, as well as the nuclear spin statistical weights g_{ns}, the initial state energy E_{i}, the transition frequency ν_{if}, the partition function $Q(T)$, and the transition moment S_{if}.

Nuclear spin statistical weights arise due to the degeneracy of nuclear energy levels. To obtain them, symmetry information is needed as well as nuclear spins. The former is readily available from our surface program (X)SURF, and nuclear spins are tabulated for all relevant isotopes. With this, the weights can be calculated using the formula by Landau and Lifshitz and group theory.[33,34] The results are integer factors that apply to transitions between states with certain symmetries and may even be zero such that transitions are not present in a spectrum.

Rovibrational energies are calculated in a two-step procedure. First, the VCI problem is solved for a non-rotating molecule. Then, a rovibrational wavefunction of the form $|\Psi_{\mathrm{rovib.}}\rangle = \sum_{rv} c_{rv} |\Phi_r\rangle|\Phi_v\rangle$

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is constructed, where the vibrational basis functions are those resulting from the preceding VCI calculation and the choice of the rotational basis follows Ref. 35. The coefficients c_{rv} are obtained by diagonalizing the Hamiltonian matrix using the complete Watson operator, where the vibrational parts are calculated numerically and the rotational parts can be determined analytically.[36,37] The vibrational basis usually consists of the ground and fundamental states as well as first overtones and combination bands but strongly depends on the density of vibrational states in the part of the spectrum of interest. Larger bases may be necessary for regions with vibrational states lying closely together due to stronger interactions. On the other hand, including higher excited states has proven to have little effect on the spectrum in the region of a vibrational mode being well separated. For a detailed spectrum, usually rotational states up to $J_{\max}$ = 30–40 are considered with no truncation of the rotational basis of $2J + 1$ basis functions.

For the partition function, the computed energy levels are used in combination with the approximate separation of rotational and vibrational contributions[38] for highly excited states. This usually gives accurate results, well within error bars of other quantities,[31,38] and allows for a separate convergence of the rotational and vibrational parts. Together with the Boltzmann factors of the lower and upper states, temperature dependence of the intensities is taken into account.

Transition moments are calculated from the dipole surfaces and rovibrational wavefunctions. Dipole surfaces are represented using an n-mode expansion in normal coordinates and polynomials for the individual terms. As with the Hamiltonian matrix elements, the integrals factorize into vibrational and rotational contributions. The rotational parts can again be solved analytically,[31,33] while the vibrational part is computed numerically.

Combining all these quantities, one finally obtains line intensities, and choosing a line shape allows us to compute the spectra. To account for Doppler broadening due to physical effects such as pressure, a Voigt profile may be applied to the line shapes. However, in this work, broadening effects have been neglected, and pure stick spectra are presented.

IV. RESULTS AND DISCUSSION

A. Potential fit

The fitting procedure described in Sec. III B is designed to provide an analytic function with as few terms as possible to reproduce the *ab initio* data with an error bar smaller than that inherent to the data points. Harmonic wavenumbers and correspondingly fundamentals have typical error bars of 5 cm^{-1} at the frozen core CCSD(T)-F12a level,[39–41] and the target accuracy for the potential fit was therefore 0.1%. This was obtained with 600 functions for configurations with energies in the range of 0–10 000 cm^{-1} above the minimum energy structure for both the fit and control sets. Plots of the fitting error against the energy above the *cis* structure are displayed in Fig. 1. Panel (a) contains the cumulative rms deviation, and a scatter plot of the error distribution is displayed in panel (b). The least squares problem was weighted to result in a constant percentage error, and 90% of the data was used for training (displayed in red) and 10% for a control set (displayed in blue). The solid black line in panel (a) represents the target cumulative rms deviation of 0.1%, and the solid black lines in panel (b) mark the 5σ error range ±0.5%.

B. Fundamental transitions

The computed fundamental vibrations of *cis*- and *trans*-HOPO are summarized in Tables I and II, respectively.

The agreement between ICPH-VCI and WH-VCI calculations for *cis*-HOPO is, in general, excellent. The only slightly larger deviation between these two calculations can be seen for ν_3 of the

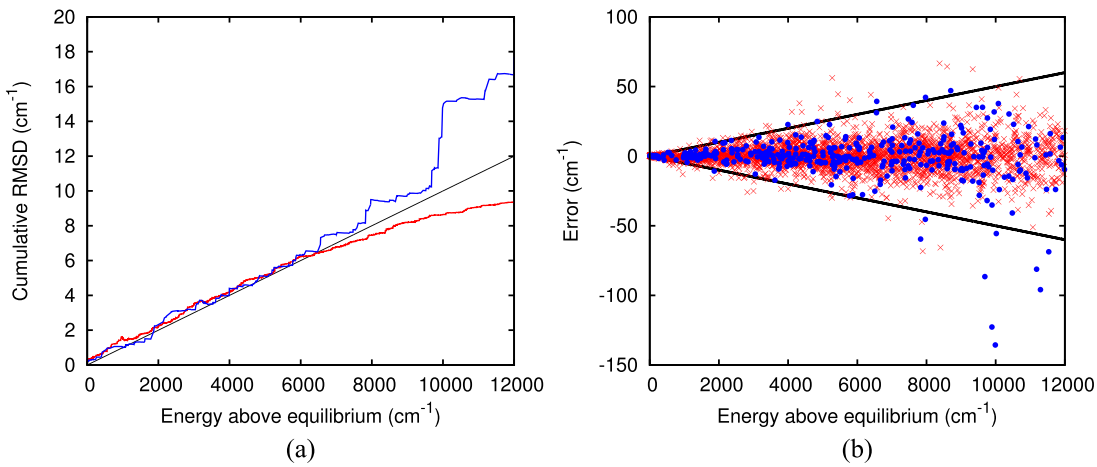

FIG. 1. (a) Cumulative rms deviations between the analytic potential and the *ab initio* data as a function of energy above the global minimum for the fit set (red) and the control set (blue). (b) rms deviations between the analytic potential and the *ab initio* data plotted against energy above the global minimum for the fit set (red) and the control set (blue).

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TABLE I. Computed (VCI) and experimental fundamental vibrations (in cm^{-1}) of *cis*-HOPO. Infrared intensities are given in km/mol.

		HOPO							DOPO				
Mode	Sym	Expt.[a]	Expt.[b]	Expt.[c]	Harm.	ICPH[d]	WH[d]	IR Int.	Expt.[c]	Harm.	ICPH[d]	WH[d]	IR Int.
1	A′		3555.1	3516.9	3753.0	3568.7	3569.2	89.7	2594.3	2729.7	2633.1	2633.1	63.1
2	A′	1258.5	1252.5	1248.2	1273.6	1258.8	1260.4	89.1	1256.5	1272.2	1260.3	1259.8	91.6
3	A′		904.4	945.4	956.7	909.9	911.4	8.2	732.6	738.0	707.5	714.6	26.8
4	A′	851.0	841.5	865.2	872.0	852.9	850.4	245.5	871.7	875.4	860.9	861.1	192.4
5	A′				397.6	390.7	393.2	45.2		364.2	360.7	361.9	44.3
6	A″		524.2	573.0	560.8	525.2	529.0	131.4		424.2	407.3	409.8	74.7

[a]Experimental gas phase value taken from Refs. 8 and 9.
[b]Experimental values measured in Ar matrix.
[c]Experimental values measured in N_2 matrix.
[d]VCI results from either the internal coordinate path Hamiltonian (ICPH) or the Watson Hamiltonian (WH).

deuterated isotopologue. However, a difference of about 7 cm^{-1} can easily result from the two different fits of the PES, and thus, we consider these values to be within the range of accuracy that can be expected from these calculations. The agreement between both theoretical approaches supports the equivalence of these methods for the calculation of the fundamental modes and that tunneling splittings are insignificant for the fundamentals. The agreement with the gas phase measurements is excellent, while deviations with respect to argon matrix isolation experiments are slightly larger. However, considering the difference between the gas phase and the matrix isolation values, matrix effects can easily account for 10 cm^{-1} and are thus in the same range or even larger than the differences or uncertainties within our calculations.

The results for *trans*-HOPO (Table II) lead to the same conclusions: both the ICPH and WH approaches are applicable for computing the fundamental transitions, tunneling splittings must be very small, and the principal reason for the deviations between the theory and the experiment are the matrix effects in the latter. Comparing the fundamental modes obtained from our ICHP-VCI and WH-VCI calculations, we observe a slightly larger difference of about 8 cm^{-1} for ν_6. This mode is dominated by the out-of-plane coordinate, which connects the two minima of the *cis*–*trans* isomerization along the reaction coordinate. Due to the shallow minimum, a PES comprising just the local minimum of the *trans* conformer is very sensitive to the fitting parameters, and thus, a slightly larger difference between both VCI methods must be expected. This has severe impact on mode ν_3, which shows a Fermi resonance with the overtone of ν_6. As the fundamental of ν_6 is sensitive to fitting parameters, its overtone is even more so and the WH-VCI calculation for ν_3 was found to be numerically unstable. Hence, we do not provide a WH-VCI value for this transition.

The problems of stability do not arise for the fundamentals of the deuterated isotopologue because the vibrational state energies are lower and thus less affected by the nearby transition state. Again, the agreement of both types of calculations is excellent, and tunneling effects on the vibrational spectrum are negligible. The agreement of the computed results with the values obtained from N_2 matrix experiments is less satisfactory than the comparison with the Ar results for *cis*-HOPO. For example, the difference of the OH-stretching mode accounts for almost 50 cm^{-1}, which indicates very strong matrix effects most likely arising from the large quadrupole moment of N_2. Note that the vibrational spectrum of *trans*-HOPO could be determined in the N_2 matrix,

TABLE II. Computed (VCI) and experimental fundamental vibrations (in cm^{-1}) of *trans*-HOPO. Infrared intensities are given in km/mol.

		HOPO					DOPO				
Mode	Sym	Expt.[a]	Harm.	ICPH[b]	WH[b]	IR Int.	Expt.[a]	Harm.	ICPH[b]	WH[b]	IR Int.
1	A′	3580.2	3807.6	3627.3	3629.9	132.9	2641.9	2771.9	2676.6	2676.9	83.3
2	A′	1263.0	1297.2	1285.3	1284.3	70.3	1262.1	1296.1	1283.1	1282.7	80.1
3	A′	932.9	956.3	942.1	...	...	705.4	707.9	699.7	702.9	17.7
4	A′	856.4	857.2	840.3	841.2	111.6	863.6	867.3	851.1	849.9	237.0
5	A′		417.9	413.8	414.2	6.2		404.0	399.0	399.8	5.1
6	A″		482.4	453.8	461.7	85.2		359.1	341.6	338.9	39.7

[a]Experimental values measured in N_2 matrix.[10]
[b]VCI results from either the internal coordinate path Hamiltonian (ICPH) or the Watson Hamiltonian (WH).

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TABLE III. VCI overtone transition energies (in cm^{-1}) and infrared intensities (in km/mol) for the out-of-plane mode of *cis*-HOPO and *cis*-DOPO.

		HOPO					DOPO				
No.	Sym.	Harm.	1D[a]	ICPH[b]	WH[c]	IR-Int.	Harm.	1D[a]	ICPH[b]	WH[c]	IR-Int.
ν_6	A″	560.8	522.2	525.2	525.3	131.1	424.1	403.9	407.3	409.2	74.7
$2\nu_6$	A′	1121.6	1011.7	1060.8	(1034.7)	6.3	848.2	787.5	812.2	808.6	5.7
$3\nu_6$	A″	1682.4	1470.2	1590.4			1272.3	1152.2	1208.6	1209.1	0.3
$4\nu_6$	A′	2243.2	1897.3	2114.4			1696.4	1498.4	1604.3	(1646.7)	0.0
$5\nu_6$	A″	2804.0	2289.4	(2371.7)			2120.5	1784.8	(1865.8)		
$6\nu_6$	A′	3364.8	2637.9	(2585.5)			2544.6	2053.3	(2145.1)		
$7\nu_6$	A″	3925.6	2937.3	(2929.1)			2968.7	2309.5	(2355.2)		

[a]VCI results from 1D calculations using the torsional internal coordinate.
[b]VCI results obtained from calculations based on the internal coordinate path Hamiltonian. Values in parentheses indicate vibrational states with an uncertain state identity due to strong couplings (see the text).
[c]VCI results obtained from calculations based on the Watson Hamiltonian. Values in parentheses indicate vibrational states, which were found to be numerically unstable with respect to either the potential parameters or the correlation space.

but not in the Ar matrix. Therefore, we consider our ICPH-VCI calculations to be best estimates for the transitions in the gas phase which will assist in the assignment of any future gas phase experimental data.

C. Frequencies of out-of-plane mode

The most interesting coordinate of the HOPO molecule is the out-of-plane torsion, which connects both minima and which is responsible for the fundamental mode at about 525.9 cm^{-1} (ICPH-VCI, *cis*-HOPO). If they are significant, tunneling effects should be seen for the vibrational states where the energy in the torsional mode approaches the barrier to isomerization. Moreover, since ν_6 of *cis*-HOPO shows very strong IR intensity, it may be possible to observe torsional overtones in sensitive experiments. We have therefore computed a series of overtones for ν_6 of *cis*-HOPO and *trans*-HOPO, and the results are summarized in Tables III and IV, respectively. Our calculations revealed that the one-dimensional potentials along the out-of-plane normal coordinate for *cis*- and *trans*-HOPO show strong quartic contributions. For *cis*-HOPO, the force constant k_{2222} was found to be 4010 cm^{-1}, while an even larger value of 5983 cm^{-1} was determined for *trans*-HOPO. For the corresponding effective potentials within the VSCF calculations, these values increased even further. This strong quartic contribution led to an enhanced sensitivity with respect to the potential parameters and to suboptimal modals, which, in turn, resulted in numerical instability in the WH-VCI calculations, and it was not possible to converge the overtones. We therefore report WH-VCI results for the fundamental transitions and the first overtones only.

The IPCH calculations use trigonometric functions for the torsional mode, and a Fourier grid for quadrature and numerical instabilities was not encountered. However, the computed ICPH-VCI overtones display an unusual pattern. In Table III, we report 1D VCI calculations in the torsional coordinate, as well as the full-dimensional ICPH-VCI values. The 1D results follow the expected pattern, where the energy spacing between levels reduces with torsional quanta. The spacing in the ICPH-VCI values is not

TABLE IV. VCI overtone transition energies (in cm^{-1}) and infrared intensities (in km/mol) for the out-of-plane mode of *trans*-HOPO and *trans*-DOPO.

		HOPO					DOPO				
No.	Sym.	Harm.	1D[a]	ICPH[b]	WH[c]	IR-Int.	Harm.	1D[a]	ICPH[b]	WH[c]	IR-Int.
ν_6	A″	482.4	423.0	453.8	461.3	86.5	359.1	295.2	341.6	338.6	40.0
$2\nu_6$	A′	964.8	825.5	861.7	(962.5)	73.5	718.2	582.7	656.9	645.2	11.2
$3\nu_6$	A″	1447.2	1203.3	1236.0			1077.3	902.7	963.6	960.3	0.0
$4\nu_6$	A′	1929.6	1549.6	(1579.9)			1436.4	1210.8	(1255.6)	(1423)	0.3
$5\nu_6$	A″	2412.0	1847.2	(1857.3)			1795.5	1497.4	(1524.1)		

[a]VCI results from 1D calculations using the torsional internal coordinate.
[b]VCI results obtained from calculations based on the internal coordinate path Hamiltonian. Values in parentheses indicate vibrational states with an uncertain state identity due to strong couplings (see the text).
[c]VCI results obtained from calculations based on the Watson Hamiltonian. Values in parentheses indicate vibrational states, which were found to be numerically unstable with respect to either the potential parameters or the correlation space.

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regular, and in particular, the 1 → 2 and 2 → 3 transition energies (535.6 cm^{-1} and 529.0 cm^{-1}) are both larger than the 0 → 1 transition (525.2 cm^{-1}). This is, in part, due to the strong quartic coupling observed in the WH-VCI calculations and also, in part, due to strong state mixing between $\nu_3 + n\nu_6$ and $(n + 2)\nu_6$, which acts to raise the energy of the torsional levels up to $5\nu_6$, above which the coupling lowers the energy of the torsional states. This leads to a small spacing of 213.8 cm^{-1} for 5 → 6, which returns to 343.6 cm^{-1} for 6 → 7, above which the states become those of a free rotor. The strong mixing of the state identities and the sensitivity of the coupling between $\nu_3 + n\nu_6$ and $(n + 2)\nu_6$ to the size of the VCI correlation space adds significant uncertainty to our predicted values for the torsional line positions above $4\nu_6$.

The pattern of the torsional states for *trans*-HOPO is much more regular than for *cis*-HOPO. There is only a small difference between the 1D-VCI, ICPH-VSCF, and ICPH-VCI transition energies, and the state identities are clean. Figures 2 and 3 show the ICPH-VSCF wavefunctions of HOPO and DOPO for the *cis*- and *trans*-isomers of both isotopologues. Based on the VSCF-ICPH wavefunctions, tunneling effects are insignificant for *cis*-HOPO states up to $4\nu_6$ and the *trans*-HOPO states up to $2\nu_6$. For the higher states, tunneling is significant and these are shown in magenta.
For the deuterated isotopologues, the situation is less problematic, and we can also use our WH calculations, which provide additional information. According to Table IV, our ICPH and WH results for *trans*-DOPO agree up to $3\nu_6$, and consequently, tunneling should be very small. This implies that the tunneling is also small for up to $5\nu_6$ for *cis*-DOPO, which is borne out by the ICPH-VSCF wavefunctions. However, the ICPH-VCI line positions are again not regular, in particular, due to strong coupling arising between $4\nu_6$ and $\nu_3 + 2\nu_6$ and between similar resonances for higher lying states of *cis*-DOPO.

In summary, tunneling appears to be of no importance for the lower transitions of HOPO and DOPO but starts to be important for high-lying overtones above 2200 cm^{-1} for *cis*-HOPO and above 1700 cm^{-1} for *cis*-DOPO. While the first overtones still bear some IR intensity, the high-lying overtones are IR inactive, and thus, tunneling effects should not be seen in experimental spectra.

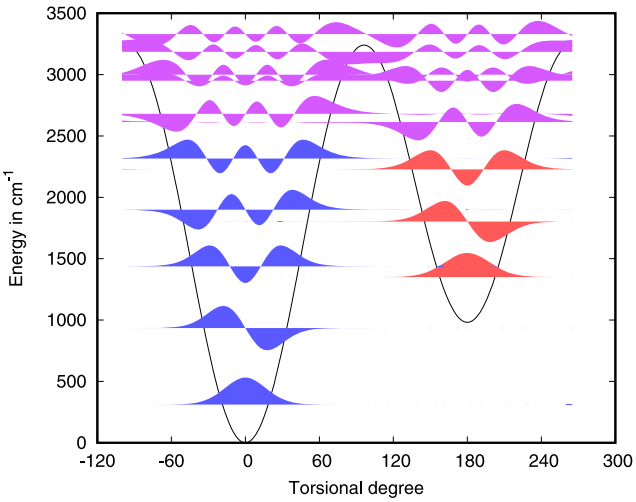

FIG. 2. Torsional wavefunctions for *trans* and *cis*-HOPO from an ICHP-VSCF calculation optimized for the zero point. States with significant tunneling character are displayed in magenta.

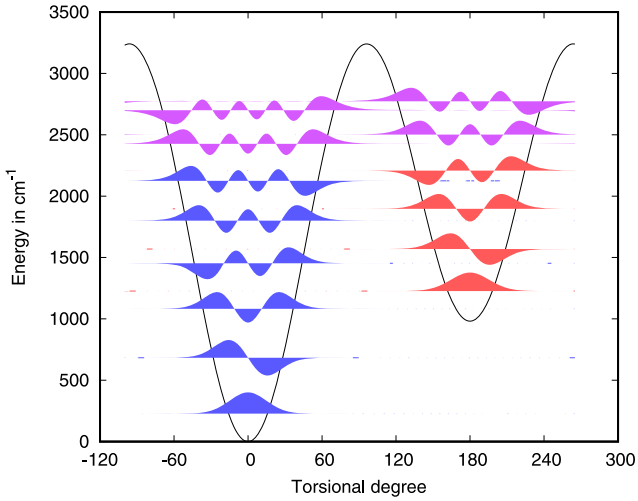

FIG. 3. Torsional wavefunctions for *trans* and *cis*-DOPO from an ICHP-VSCF calculation optimized for the zero point. States with significant tunneling character are displayed in magenta.

D. Rovibrational spectrum of the out-of-plane mode

The rovibrational infrared spectrum of the OOP mode ν_6 was calculated, as described in Sec. III D. Potential energy surfaces were calculated at the CCSD(T)-F12a/cc-pVTZ-F12 level of theory for 1D and 2D terms and using a cc-pVDZ-F12 basis for 3D and 4D terms of the *n*-mode representation. For the dipole surfaces, analytic gradients implemented in MOLPRO for the CCSD(T) method were used again up to 4D terms.

For the VCI calculations, a moderate configuration space was used with maximum excitations of 5 and a maximum sum of all vibrational quanta of 12. This results in frequencies and CI coefficients differing from the more elaborate calculations done for the previous vibrational transitions. Notably, the band center is shifted to a higher frequency, as can be expected, but the overall impact on the rovibrational spectrum is only minor. Beyond the ground state and fundamentals, first overtones and combination bands were calculated to have a sufficiently large vibrational basis for the subsequent rovibrational calculations. Considering only such states for which the (harmonically approximated) vibrational energy is below a threshold of 5000 cm^{-1} leads to 26 vibrational states taken into account. Rotational excitations were examined up to a value of J_{max} = 35. This results in a total of 33 670 rovibrational energy levels.

The nuclear spin statistical weights for HOPO are $g_{ns} = \{4, 4\}$ for $\{A', A''\}$ states, respectively. This means that the complete spectrum is simply scaled by a factor of 4. The partition function was determined using the calculated energy levels as well as the aforementioned rotation–vibration–separation approximation for higher

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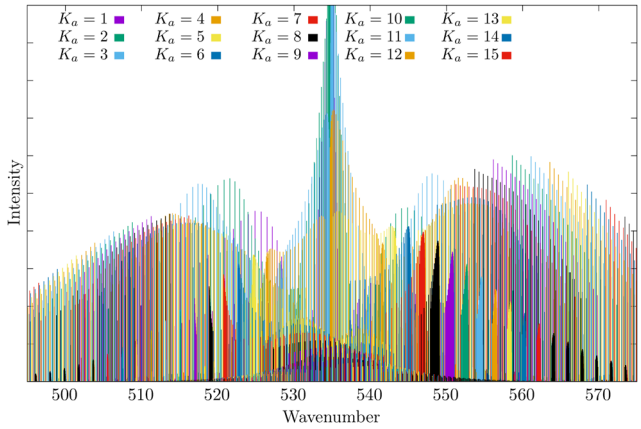

FIG. 4. Rovibrational spectrum of the OOP mode of -HOPO. Transitions are distinguished by their upper level quantum number K_a up to K_a = 15 and shown in black above.

excited states. For a temperature of T = 300 K, it was determined to be Q(300 K) = 85 305.

Filtering for transitions with a significant intensity results in about 13 000 transitions in the region of the vibrational OOP mode with an approximate quantum number of ν_6. The spectrum for a temperature of 300 K is provided in Fig. 4. It shows a dominant A type band structure with a gathered Q branch. An asymmetry between the P- and R-branches is present with a wider P-branch and a comparatively squeezed R-branch. Color coding is used to distinguish lines with a specific upper level quantum number K_a, allowing for an easier analysis of sub-features and progressions in the spectrum. These can be seen well in parts of the spectrum such that band edges and decreasing and increasing energy spacing become apparent.

V. SUMMARY AND CONCLUSION

Multidimensional potential energy surfaces using different coordinate systems have been used in subsequent VSCF and VCI calculations, relying on the internal coordinate path Hamiltonian or the standard Watson Hamiltonian. The agreement of both approaches for the fundamental modes of this system indicates that tunneling effects are without relevance for these low-lying transitions. However, for high-lying overtones of the torsional mode, the situation changes and tunneling becomes effective which, in turn, has impact on the reaction barrier of the *cis–trans* isomerization. The rovibrational spectrum of the *cis*-HOPO OOP mode has been studied by high-level *ab initio* calculations, including vibrational overtones and combination bands and considering J-values as large as 35. Strong rotational sidebands are apparent, featuring asymmetric P- and R-branches and a prominent Q-branch. Experimental gas phase measurements would be highly desirable for these species in order to confirm the assignments and findings of this study.

SUPPLEMENTARY MATERIAL

See the supplementary material for geometrical parameters of the stationary points and a detailed analysis of the rovibrational spectrum given in Fig. 4. A Fortran subroutine with the analytic potential is provided.

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The data that support the findings of this study are available from the corresponding author upon reasonable request.

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